

Amitej Singh Datta

amitejsingh5@gmail.com | +65 90555056 | linkedin.com/amitej-singh | github.com/AmitejSingh1 | amitejsingh-datta.vercel.app

Professional Summary

AI/ML focused Computer Science graduate currently pursuing an M.S. in Data Science & Machine Learning at NUS, specializing in computer vision and medical image segmentation. Published first-author work (ICCIS 2025, Springer) on a hybrid deep learning architecture for prostate segmentation, and hold a published patent for a multimodal sign language translation system. Proficient in PyTorch, TensorFlow, OpenCV, and Hugging Face across the full model lifecycle. Known for methodical experimentation, clear technical communication, and following problems through to a working, well-benchmarked solution.

Core Competencies

- **AI & ML:** Deep Learning, CNNs, Transformers, LoRA/PEFT, RAG
- **Computer Vision:** Segmentation, Pose/Object Detection, YOLO, ViT, OpenCV, MediaPipe
- **Tools:** PyTorch, TensorFlow, Hugging Face, FastAPI, Flask, Git
- **Languages:** Python, C++, Java, JavaScript, SQL

Education

Vellore Institute of Technology, Chennai B.Tech, Computer Science & Engineering (AI & ML)	Sep 2022 – July 2026 GPA: 9.09
National University of Singapore M.S., Data Science & Machine Learning	Aug 2026 – Present

Experience

Software Intern · Casamed HealthTech (Remote)	Oct 2024 – Jan 2025
• Automated extraction and analysis of key metrics from medical reports by building Python-based OCR and parsing pipelines, cutting the manual review workload for clinical staff and standardising data intake across the platform. • Engineered pose-detection scripts in OpenCV and MediaPipe alongside the product team to power AI-assisted physiotherapy assessments, improving backend ML model-integration reliability and data flow into the platform.	
Student Trainee Intern · DRDO – CAIR, Bengaluru	May 2025 – Jul 2025
• Built a video summarisation pipeline using ResNet-50 for frame-level feature extraction and unsupervised clustering (K-Means, DBSCAN, motion-based segmentation), automatically identifying keyframes to cut manual review time on long-format video. • Benchmarked the system against TVSum ground-truth annotations (Precision, Recall, F1-score) and deployed it end-to-end via a Flask-React web application, delivering a working upload-to-summary pipeline validated against a standard research benchmark.	

Projects

LoRA Fine-tuning Framework for Medical Image Segmentation	github/MedSAM_LoRA
• Implemented a from-scratch LoRA (Low-Rank Adaptation) trainer applied to MedSAM (ViT-B) for prostate segmentation, achieving 95.7% Dice / 91.9% IoU and surpassing the VGG-UNet baseline with 82% fewer trainable parameters. • Optimised training via image embedding caching, cutting per-epoch time from 60 minutes to 3 minutes (a 60% overall speedup) while maintaining state-of-the-art segmentation performance.	
AI-based Prostate Segmentation on Micro-Ultrasound Images	github/ProstateSegmentation
• Designed a hybrid VGG16-UNet architecture to automate prostate volume delineation on micro-ultrasound scans, replacing a labour-intensive manual radiologist workflow. • Refined the model through data augmentation, hyperparameter tuning, and post-processing boundary correction, producing clinically precise segmentation outputs validated through peer review at ICCIS 2025.	
AccessAudio: Multimodal Sign Language Translation System — Patent Published	github/AccessAudio
• Built a browser-based, real-time multimodal translation system combining ISL/ASL gesture recognition (LSTM+Transformer) with facial emotion recognition (MobileNetV2), reaching 73.19% emotion-recognition accuracy to preserve tone alongside sign content. • Engineered low-latency gesture, audio, and video transmission over WebRTC for fully browser-native, install-free deployment, resulting in a published patent for the underlying system.	

Publications & Patents

- **Hybrid VGG16-UNet for Prostate Segmentation on Micro-Ultrasound Images**, ICCIS 2025, Springer Book Series (First Author)
- **Patent Published:** Multimodal Sign Language Translation System (AccessAudio) — Application No. 202641089347

Extracurricular

- **Culture Lead**, Fraternity of Young Innovators, VIT Chennai
- **E-Summit VIP GuestCare Head**, Voyage (college tech summit)
- **Volunteer**, Khalsa Aid (Punjab flood relief operations) and **English Tutor** at SOS Children Villages of India.